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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 3: Water Jug Problem

PROGRAM:

practical lab 3.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 3.py (3.14.5)

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```
from collections import deque

capacity_a = 4
capacity_b = 3
target = 2

start = (0, 0)
queue = deque([(start, [start])])
visited = {start}
solution = None

while queue:
    (a, b), path = queue.popleft()

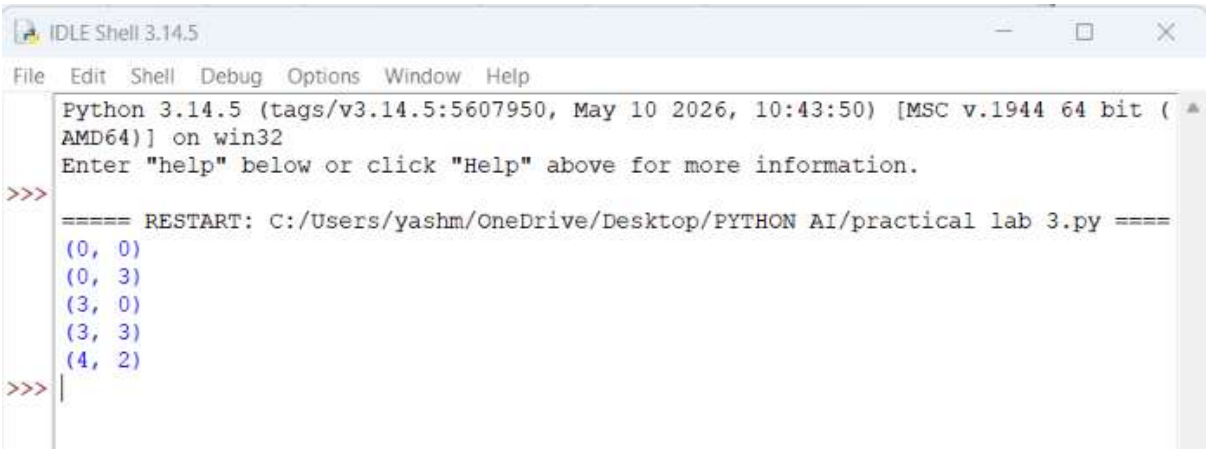
    if a == target or b == target:
        solution = path
        break

    states = [
        (capacity_a, b),                    # Fill A
        (a, capacity_b),                    # Fill B
        (0, b),                             # Empty A
        (a, 0),                             # Empty B
        (a - min(a, capacity_b - b),
         b + min(a, capacity_b - b)),       # Pour A to B
        (a + min(b, capacity_a - a),
         b - min(b, capacity_a - a))        # Pour B to A
    ]

    for state in states:
        if state not in visited:
            visited.add(state)
            queue.append((state, path + [state]))

if solution:
    for step in solution:
        print(step)
else:
    print("No solution found")
```

OUTPUT:


IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 3.py ====
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>